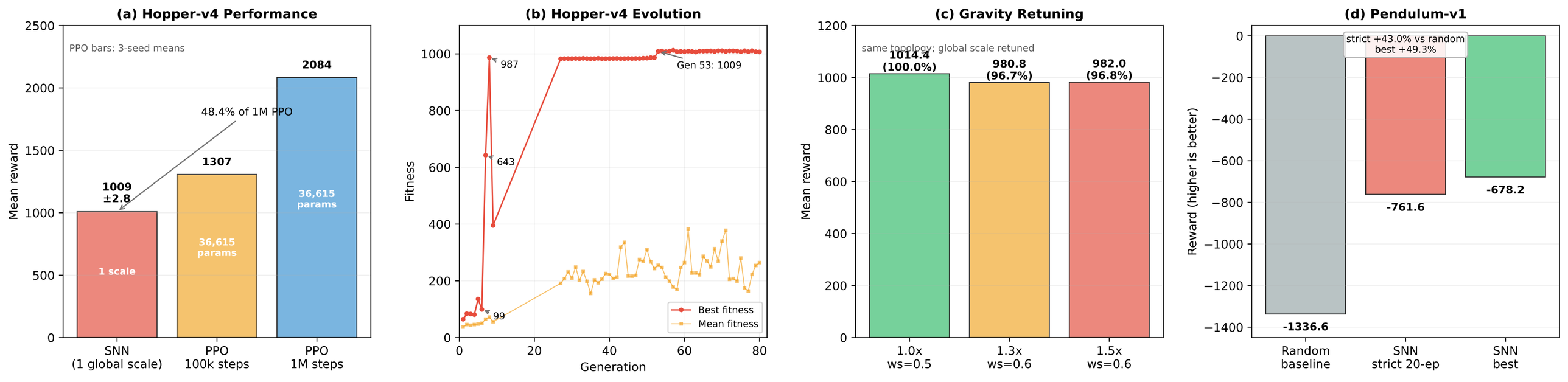
(a) Hopper-v4 Performance
PPO bars: 3-seed means
Mean reward
48.4% of 1M PPO
1009
±2.8
1307
2084
1 scale
36,615
params
36,615
params
SNN
(1 global scale)
PPO
100k steps
PPO
1M steps
(b) Hopper-v4 Evolution
Fitness
987
643
99
Gen 53: 1009
Best fitness
Mean fitness
Generation
(c) Gravity Retuning
same topology; global scale retuned
Mean reward
1014.4
(100.0%)
980.8
(96.7%)
982.0
(96.8%)
1.0x
ws=0.5
1.3x
ws=0.6
1.5x
ws=0.6
(d) Pendulum-v1
strict +43.0% vs random
best +49.3%
Reward (higher is better)
-1336.6
-761.6
-678.2
Random
baseline
SNN
strict 20-ep
SNN
best